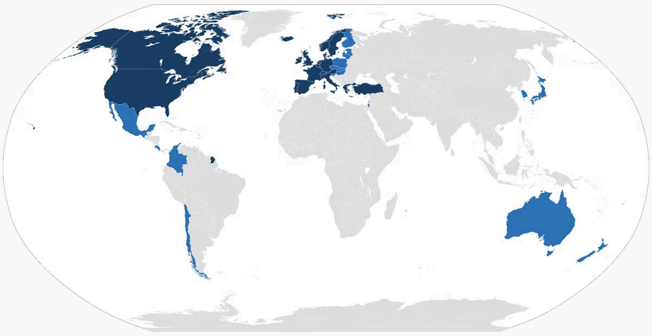


The Gender Conversion Gap: From STEM and Digital Participation to Labour-Market Power

ANALYTICAL REPORT | OECD CROSS-COUNTRY ANALYSIS | ESTONIA POLICY BRIEF

An analytical report examining whether stronger STEM and digital participation indicators are associated with stronger labour-market power outcomes across OECD countries — with a focus on Estonia's mixed benchmark profile and the policy implications of the conversion layer between education and labour-market power.

Executive Summary

This report examines whether stronger women's STEM and digital participation indicators are associated with stronger labour-market power outcomes across OECD countries. The analysis uses latest-available OECD-member country data across six indicators covering education, digital skills, labour-market access, pay, work structure, and leadership representation. The findings are framed for Estonia-facing gender-equality stakeholders and other actors interested in moving beyond participation metrics toward labour-market outcome accountability.

1

Hypothesis Results

The results do not support a simple pipeline story in which stronger representation in STEM education automatically corresponds to stronger labour-market equality. **H1** shows partial support: countries with higher shares of women among tertiary STEM graduates tend to have higher women's representation in senior/middle management, but the result is sensitive to influential observations. **H2** is not supported: higher women's STEM graduate share is not clearly associated with lower median gender wage gap indicator values. **H3** is exploratory and points in the opposite direction from expectation, while **H4** is not supported as a simple country-level association.

2

Estonia's Conversion Problem

Estonia illustrates the practical importance of this mismatch. Estonia is above the OECD median in women's share among tertiary STEM graduates and has relatively favourable gaps in youth coding, labour-force participation, and part-time employment. However, Estonia's median gender wage gap indicator is substantially above the OECD median, and women's representation in senior/middle management is below the OECD median. This suggests that participation gains do not necessarily convert into pay equality and leadership representation.

3

Main Recommendation

The main recommendation is to shift from pipeline-only monitoring to conversion-layer accountability. Estonia-facing stakeholders should track what happens after STEM education: transition into STEM-intensive employment, retention, promotion, senior/middle management representation, and pay outcomes. The recommendations in this report are evidence-informed priorities for monitoring and intervention, not causal claims proven by the cross-country tests.

> All findings in this report are cross-country associations, not causal effects. Recommendations are evidence-informed priorities for monitoring and intervention.

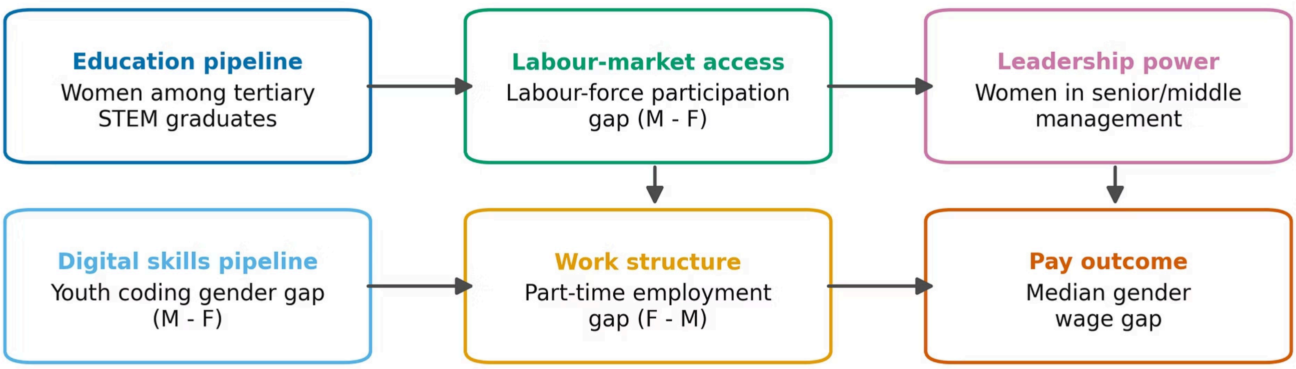
1. Problem and Stakeholder Framing

Gender-equality work often treats education and skills as the beginning of the solution: if more women enter STEM fields or gain digital skills, labour-market equality should improve later. This project examines whether that assumption is visible in country-level OECD indicators. The core question is not whether education matters, but whether pipeline-level gains reliably correspond to pay equality and leadership representation.

The report frames this issue as a “gender conversion gap”: a possible disconnect between women’s participation in STEM and digital pipelines and their later labour-market power, measured through pay and leadership indicators. The term is a framing device, not a literal cohort-level transition claim. The data do not follow the same women from graduation into careers. Instead, the analysis tests whether countries with more favourable pipeline indicators also tend to have more favourable labour-market and leadership outcomes.

The stakeholder problem is practical. If pipeline indicators are not enough, Estonia-facing gender-equality stakeholders need to know where to focus: education access, digital skills, employment entry, retention, promotion, pay systems, or leadership pathways. A monitoring strategy that stops at graduation rates or enrolment figures risks missing the mechanisms that generate persistent pay and representation gaps.

Project framing: from participation indicators to labour-market power indicators



Country-level latest-available indicators; associations, not individual cohort transitions or causal effects

The Pipeline Assumption

More women in STEM education → improved digital skills → stronger labour-market equality. Policy attention concentrates on enrolment, graduation, and access metrics.

The Conversion Gap Challenge

Pipeline gains do not automatically translate into pay equality or leadership representation. The critical question is what happens *after* graduation: employment entry, retention, promotion, and pay progression.

2. Research Question and Hypotheses

Research question: Do OECD countries with stronger women's STEM and digital participation indicators also show stronger labour-market and leadership outcomes? The project tests four working hypotheses: two primary hypotheses focused on the STEM-to-power relationship, and two secondary/exploratory hypotheses focused on adjacent mechanisms in the conversion pathway.

1

H1 — STEM-to-Leadership Association *(Primary)*

Countries with higher shares of women among tertiary STEM graduates tend to have higher shares of women in senior/middle management. This hypothesis tests whether STEM education representation is associated with leadership representation at the country level.

2

H2 — STEM-to-Pay Equality *(Primary)*

Countries with higher shares of women among tertiary STEM graduates tend to have lower median gender wage gap indicator values. This hypothesis tests whether stronger representation in the STEM education pipeline is associated with narrower pay gaps.

3

H3 — Digital Skills & Labour-Market Participation *(Exploratory)*

Countries with larger male advantages in youth coding activity tend to have larger gender gaps in labour-force participation. This hypothesis is treated as exploratory because the indicators refer to different age groups and partly different years: youth coding covers ages 16–24, while labour-force participation covers the broader working-age population.

4

H4 — Work-Structure Bottleneck *(Secondary)*

Countries where women are more concentrated in part-time employment relative to men tend to have larger median gender wage gap indicator values. This hypothesis tests whether gendered work-structure concentration is associated with pay inequality at the country level.

H1 and H2 are the **primary hypotheses** because they directly test whether the STEM education pipeline is associated with leadership and pay outcomes. Their p-values are treated as confirmatory and adjusted using the Holm procedure. H3 and H4 are **secondary/exploratory** because they test adjacent mechanisms rather than the central STEM-to-power relationship.

3. Data Source, Sample, and Indicator Definitions

The analysis uses processed OECD gender-equality indicators collected through OECD data tools and APIs. The main analysis sample contains 38 OECD member countries. Aggregate areas such as OECD, EU27, G7, and EU28 are excluded from hypothesis tests because they are not independent country observations.

The dataset uses latest-available country-level values rather than a synchronised same-year cross-section. This choice increases country coverage, but it also creates year-alignment limitations, especially for H3, where the youth coding indicator and the labour-force participation indicator are not always observed in the same year.

Indicator	Definition	Direction	Role in Analysis	Coverage (n)
Women among tertiary STEM graduates	Share of women in all STEM tertiary graduates (%)	Higher = stronger women's representation	Pipeline / H1, H2	38
Youth coding gender gap	Male minus female coding activity, pp	Higher = larger male advantage	Digital skills / H3	32
Labour-force participation gap	Male minus female LFP rate, pp	Higher = larger gender participation gap	Labour access / H3	38
Median gender wage gap indicator	Median earnings gap (%), broad measure	Higher = larger wage inequality	Pay outcome / H2, H4	38
Part-time employment gap	Female minus male part-time share, pp	Higher = women are more concentrated in part-time employment relative to men	Work structure / H4	36
Women in senior/middle management	Share of women in senior and middle management roles (%)	Higher = stronger leadership representation	Leadership outcome / H1	30

Indicator coverage is sufficient for the planned country-level analysis. STEM graduate share, labour-force participation gap, and median gender wage gap are available for all 38 OECD member countries. Youth coding has 32 OECD observations, part-time employment gap has 36, and women in senior/middle management has 30. These data are sufficient for descriptive benchmarking and hypothesis testing at the country level, but not for causal claims or individual career-transition analysis.

4. Descriptive Findings: Estonia's Mixed Benchmark Profile

Estonia's benchmark profile is mixed rather than uniformly strong or weak. On several pipeline and labour-market access indicators, Estonia performs more favourably than the OECD median. However, on the two labour-market power outcomes — the median gender wage gap indicator and women's representation in senior/middle management — Estonia's position is less favourable. This mismatch is the descriptive core of the report and directly motivates the conversion-gap framing.

Three descriptive findings are especially important:

41.55%

Women in STEM Graduates

Above OECD median of 34.25% — Estonia's strongest pipeline indicator.

19.77%

Median Gender Wage Gap

Substantially above OECD median of 10.74% — Estonia's largest outcome gap.

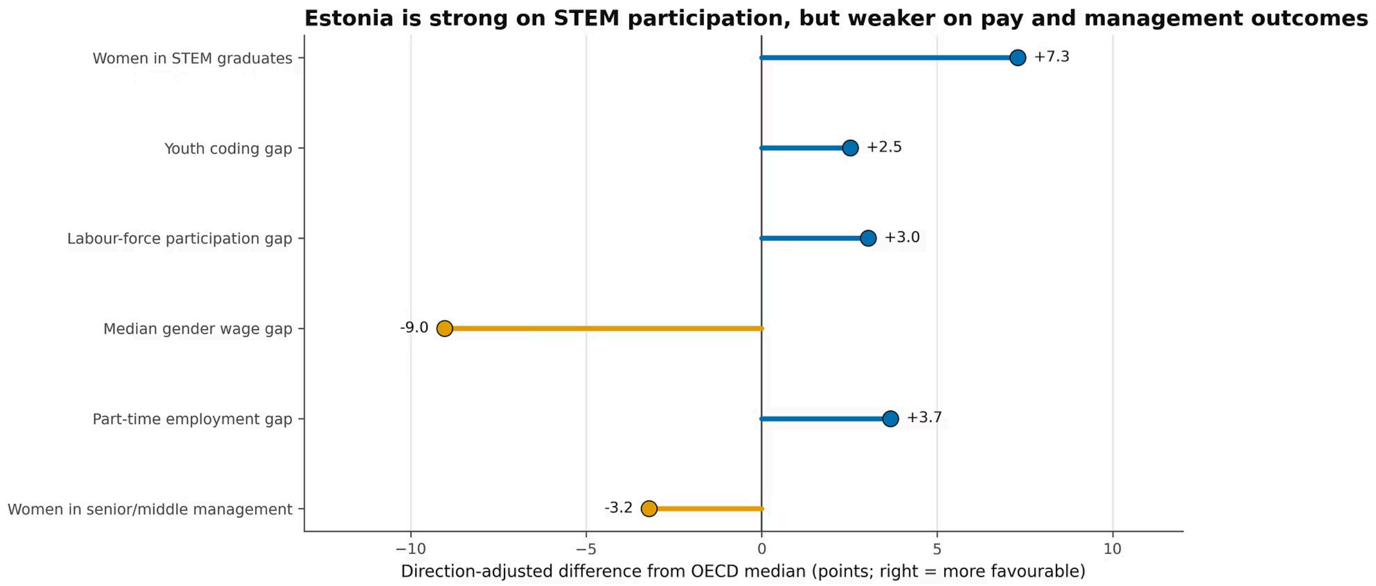
31.51%

Women in Management

Below OECD median of 34.72% — underperformance on leadership representation.

Indicator	Estonia	OECD Median	Estonia vs Median
Women among tertiary STEM graduates	41.55%	34.25%	▲ More favourable
Youth coding gap (male – female, pp)	7.52 pp	10.05 pp	▲ More favourable
Labour-force participation gap (male – female, pp)	4.92 pp	7.97 pp	▲ More favourable
Part-time employment gap (female – male, pp)	5.54 pp	9.21 pp	▲ More favourable
Median gender wage gap indicator	19.77%	10.74%	▼ Less favourable
Women in senior/middle management	31.51%	34.72%	▼ Less favourable

The figure below shows the same comparison using direction-adjusted differences from the OECD median, where positive values indicate a more favourable position for Estonia. The pattern is clear: Estonia performs well on STEM representation and several access-related indicators, but less favourably on pay and leadership outcomes. Estonia's challenge is therefore not only about getting women into STEM education. The more important question is whether education and skills convert into equitable pay, career progression, and leadership representation.



5. Hypothesis Testing Methods

Each hypothesis is tested as a cross-country association between two continuous country-level indicators. The analysis reports the pairwise complete sample size, Pearson correlation for linear association, Spearman correlation as a robustness check for monotonic association, simple OLS regression to express the association in original indicator units, and pairwise year-gap diagnostics.

Primary Test Procedure

For H1 and H2, Pearson correlation p-values are treated as the primary confirmatory tests because these two hypotheses form the pre-specified primary family: the relationship between women's STEM graduate representation and labour-market power outcomes. Their p-values are adjusted using the Holm procedure as a multiple-testing check. Spearman correlations serve as robustness checks.

Secondary/Exploratory Tests

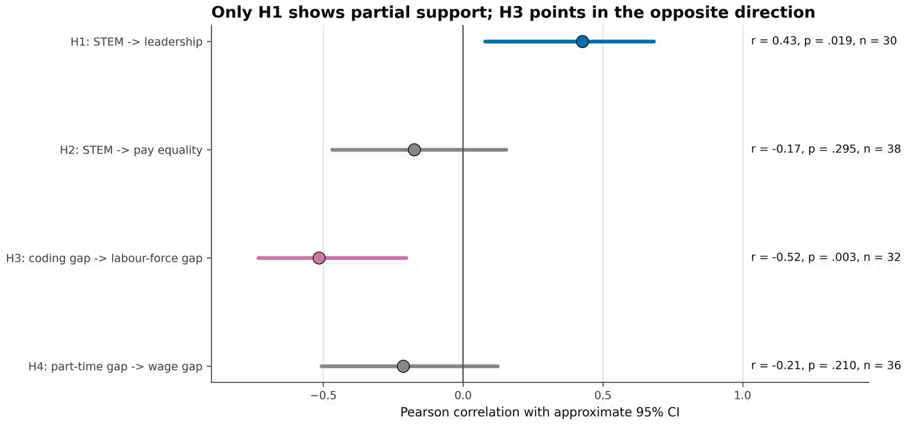
H3 and H4 are interpreted as secondary/exploratory tests. Their p-values are reported and discussed, but they are not used for strong confirmatory claims. The year-gap diagnostic is especially important for H3, where youth coding and labour-force participation values can be several years apart for some countries.

Simple OLS regression is used to express the same bivariate relationship in original indicator units. In a single-predictor bivariate model, the OLS slope test and Pearson correlation test evaluate the same linear association. Therefore, OLS is interpreted as an effect-size estimate rather than as an independent confirmation of the Pearson result.

> All findings are cross-country associations, not causal effects. OLS slope estimates describe the observed country-level association in original units; they do not imply that changing one indicator would mechanically cause another indicator to change.

6. Results: The Simple Pipeline Story Is Not Strongly Supported

Across the four hypotheses, the results challenge a simple linear story in which stronger STEM or digital participation automatically corresponds to stronger pay and leadership outcomes. The primary hypothesis family receives only partial support, and the secondary/exploratory tests do not provide a simple alternative mechanism.



H1 — Partial Support

H1 shows a moderate positive linear association between women's STEM graduate share and women's representation in senior/middle management. The result remains significant after Holm adjustment within the primary hypothesis family. However, the rank-based association is weaker, and influence diagnostics show that the result is sensitive to Japan. This is interpreted as partial evidence, not as a robust OECD-wide pattern.

H2 — Not Supported

H2 does not show a meaningful association between women's STEM graduate share and the median gender wage gap indicator. The direction is weakly negative as expected, but the relationship is not statistically significant. This means that countries with higher women's STEM graduate share do not consistently show lower median gender wage gap indicator values in this dataset.

H3 — Exploratory, Opposite Direction

H3 shows a negative association, opposite to the expected positive direction. Because the indicators refer to different age groups and partly different years, this result is treated as exploratory. It suggests that youth digital participation gaps and broad labour-market participation gaps should not be interpreted as the same mechanism.

H4 — Not Supported

H4 does not show a meaningful association between the part-time employment gap and the median gender wage gap indicator. This does not prove that part-time work is irrelevant to pay inequality. It only shows that this country-level bivariate relationship is not a strong standalone marker of the median wage gap indicator in the current dataset.

Hypothesis	Pearson r	Raw p	Holm p	Spearman ρ	n	Verdict
H1 STEM → Management	+0.43	0.019	0.037	+0.25	30	Partial support; sensitive to Japan
H2 STEM → Pay Gap	−0.17	0.295	0.295	−0.12	38	Not supported
H3 Coding → LFP Gap	−0.52	0.003	n/a	−0.46	32	Exploratory; opposite direction
H4 Part-time → Pay Gap	−0.21	0.210	n/a	−0.21	36	Not supported

7. Interpretation: Why the Expected Story May Not Hold

The non-confirmed and partially confirmed hypotheses are not a project failure. They are the main substantive insight. A simple pipeline model assumes that better representation in education or digital skills should later appear in pay and leadership outcomes. The results suggest that this assumption is too simple.

The cross-country tests cannot identify the mechanisms behind the mismatch, but they are compatible with several explanations operating at the same time. These mechanisms should be treated as interpretation and future investigation priorities, not as causal conclusions directly proven by the current dataset.

→ Entry Gap

Women may graduate in STEM fields but not enter STEM-intensive jobs at the same rate as men. Field of study and sector of employment are not the same indicator, and transition rates from STEM education into STEM employment may differ by gender.

→ Retention and Progression Gap

Women may enter technical or high-skill jobs but leave earlier, progress more slowly, or face different promotion conditions. Promotion and pay systems may reward availability, bargaining, informal networks, or seniority patterns unevenly.

→ Structural Sorting

Women may be concentrated in lower-paid firms, sectors, occupations, or roles even within STEM-related employment. Country-level education indicators cannot show whether women and men with similar qualifications are entering the same types of jobs, firms, or career tracks.

→ Care and Career Interruptions

Care responsibilities and career breaks can affect wage growth and promotion timing. This is especially relevant for Estonia, where OECD research highlights that women have high employment and education levels, but are still less likely to reach top positions, and that career breaks around childbirth contribute to the still considerable gender wage gap (OECD, 2022).

> External OECD research supports this interpretation. OECD work on Estonia points to the importance of firms, bargaining power, pay transparency, and work-life balance policies in understanding the gender pay gap. Broader OECD skills research also argues that skills do not automatically translate into equal labour-market opportunities unless they are recognised, used, and rewarded fairly (OECD, 2025c).

The right response is therefore not to abandon STEM or digital-skills initiatives. It is to connect them to downstream accountability: employment entry, retention, promotion, pay, and leadership. The conversion layer is where monitoring systems and intervention design need to become more precise.

8. Recommendations: Build Conversion-Layer Accountability

The recommendations below are evidence-informed priorities for Estonia-facing stakeholders. They are not presented as causal effects proven by this analysis. Each recommendation links a statistical finding to a practical monitoring or intervention path. Together, they represent a shift from pipeline-only monitoring toward a broader accountability framework that follows the journey from education to labour-market power.

1

Track STEM-to-Work Conversion, Not Only Graduation

Estonia's high women's STEM graduate share does not align with equally strong pay and leadership outcomes. This suggests that monitoring should not stop at education indicators. Estonia-facing stakeholders should build a regular STEM-to-labour-market conversion dashboard that tracks whether women move from STEM education into STEM-intensive employment, remain in those sectors, progress into leadership, and receive equitable pay.
Key metrics: STEM graduates by gender; first jobs in STEM-intensive sectors; retention after 1, 3, and 5 years; women's share in senior/middle management; gender wage gaps by sector, occupation, and seniority.

2

Pair Leadership Programmes with Promotion Transparency

H1 provides partial evidence that women's STEM representation and women's management representation can move together, but the result is not robust enough to assume automatic conversion. Leadership interventions should therefore focus not only on individual support, but also on the systems that shape advancement.

Recommended actions include sponsorship programmes, transparent promotion criteria, leadership pipeline programmes, and regular promotion audits in STEM-intensive sectors.

Key metrics: promotion rates by gender; time to promotion; retention after first STEM job; women's share in senior/middle management; participation and outcomes in sponsorship or leadership programmes.

3

Treat Pay Equality as a Separate Policy Problem

H2 is not supported: higher women's STEM graduate share is not strongly associated with lower median gender wage gap indicator values across OECD countries. This means that pay inequality should not be treated as something that will automatically improve once women's STEM representation increases.

Estonia-facing stakeholders should prioritise pay-transparency tools and regular gender pay audits, especially in high-wage and STEM-intensive sectors.

Key metrics: unadjusted and adjusted gender wage gaps; pay bands; bonus gaps; occupational sorting; firm-level wage gaps; pay progression by tenure, role, and sector.

4

Separate Youth Digital Participation from Labour-Market Access

H3 shows that youth coding indicators and overall labour-force participation gaps do not align in the expected way. Digital-skills participation should therefore be monitored as an early-stage pipeline indicator, not as a direct proxy for broad labour-market equality.

Digital-skills programmes should track whether youth coding participation converts into ICT/STEM study, internships, first jobs, and longer-term labour-market outcomes.

Key metrics: coding participation by gender; ICT/STEM enrolment; internship participation; first jobs in ICT/STEM; transition rates from digital-skills programmes into employment.

5

Broaden Work-Structure Monitoring Beyond Part-Time Employment

H4 is not supported as a simple country-level relationship between the part-time employment gap and the median gender wage gap indicator. This does not mean that work structure is irrelevant. It means that part-time employment alone is too narrow a proxy for the mechanisms that may shape pay inequality.

Monitoring should include care responsibilities, parental leave uptake, career breaks, job quality, return-to-work outcomes, and promotion or pay progression after leave.

Key metrics: part-time employment by gender; parental leave uptake by gender; childcare access; career-break duration; return-to-work outcomes; post-leave promotion and pay progression.

9. Limitations

This report should be interpreted with several methodological and data limitations in mind. These limitations do not invalidate the analysis, but they define the boundary between what the results can and cannot claim. The findings are useful for identifying monitoring priorities and evidence-informed recommendations, but they should not be interpreted as causal policy effects or predictions about individual career trajectories.

Associations, Not Causal Effects

Country-level correlations and OLS regression coefficients describe associations across OECD member countries. They do not show that changing one indicator would directly cause a change in another. No causal inference framework is applied.

Year-Alignment Limitations

The dataset uses latest-available values rather than a synchronised same-year cross-section. This improves country coverage, but it creates timing limitations. H3 requires the most caution because youth coding and labour-force participation values can be several years apart for some countries.

No Individual or Cohort Tracking

The dataset does not follow individual women or cohorts from STEM graduation into employment, pay, or leadership outcomes. The analysis therefore cannot directly test whether women who graduate from STEM later enter management, remain in STEM-intensive sectors, or experience specific pay trajectories.

Unmeasured Mechanisms

Several important mechanisms are not captured in the six indicators. The data do not include firm-level pay practices, occupation, tenure, parental status, career breaks, discrimination, bargaining power, or promotion processes. These mechanisms are important for interpreting gender pay and leadership gaps, but they cannot be directly tested with the current country-level dataset.

Indicator Scope

Leadership is measured using women's share in senior/middle management, not board representation or executive decision-making power. The median gender wage gap indicator is a broad measure and should not be interpreted as equal pay for equal work. It may reflect hours, occupation, sector, seniority, career interruptions, and other structural factors.

10. Future Data Needs

A stronger future analysis would use more detailed data capable of observing the conversion pathway directly. Country-level cross-sectional associations can identify patterns and motivate hypotheses, but they cannot resolve where exactly the conversion gap occurs for women in Estonia. Future work should therefore prioritise data that connects education, employment, pay, progression, and leadership outcomes over time.

Individual and Cohort-Level Data

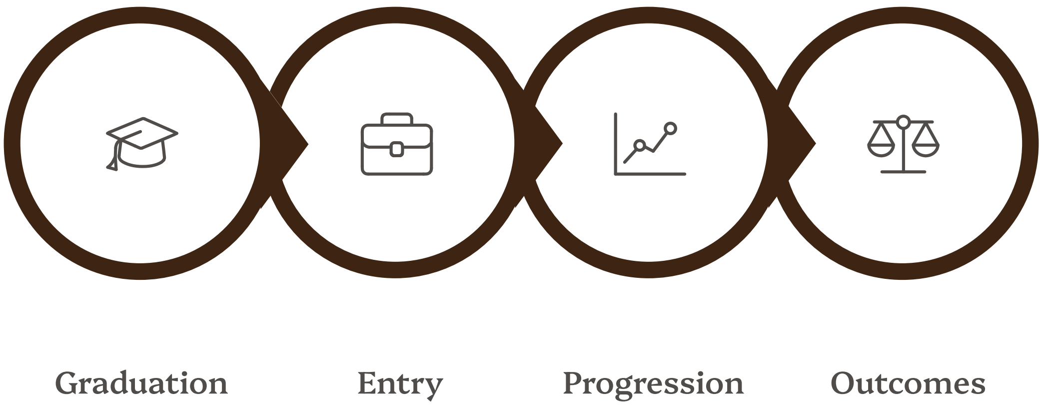
Future analysis would benefit from data linking field of study to employment outcomes at the individual or cohort level. This would make it possible to track STEM employment entry, retention, promotion timing, management entry, pay progression, parental leave, care responsibilities, and career breaks by gender.

Structural and Firm-Level Data

More granular labour-market data would also be needed to test workplace mechanisms directly. Relevant variables include sector, occupation, firm, role, tenure, hours, pay bands, bonuses, promotion decisions, and seniority. These data would help distinguish whether the conversion gap appears at labour-market entry, retention, promotion, pay progression, or leadership selection.

Programme and Intervention Data

If Estonia-facing stakeholders implement STEM, digital-skills, mentoring, sponsorship, or pay-transparency interventions, future evaluation should collect programme-level data. This would make it possible to measure not only participation, but also downstream outcomes such as internships, first jobs, retention, promotion, and pay progression.



The diagram above maps the four conversion-pathway stages that future data collection should target. Each stage represents a point where the gap between pipeline participation and labour-market power may emerge or widen, and where targeted monitoring could identify the most effective intervention points for Estonia-facing stakeholders.

Conclusion

This report does not find evidence for a simple pipeline story in which stronger women's representation in STEM and digital participation automatically translates into stronger labour-market power outcomes. Instead, the main finding is the mismatch itself: across OECD country-level data, pipeline indicators do not reliably predict pay equality or leadership representation.

Estonia makes this mismatch especially visible. It performs strongly on women's share among tertiary STEM graduates and several participation indicators, but still has a high median gender wage gap indicator and below-median women's representation in senior/middle management. This suggests that Estonia's gender-equality challenge is not only at the starting line of education and skills, but also in the conversion layer between education and labour-market power.

The policy direction is therefore to continue supporting women in STEM and digital skills, while measuring what happens next: entry into STEM-intensive work, retention, promotion, pay progression, and leadership representation. For Estonia-facing stakeholders, conversion-layer accountability offers a more useful monitoring framework than pipeline metrics alone.

Source Notes

The OECD Dashboard on Gender Gaps is the primary data source for the six country-level indicators used in the analysis. The remaining OECD publications are used for contextual interpretation and recommendation framing, especially around Estonia's gender wage gap, skills-to-labour-market conversion, paid and unpaid work, care responsibilities, and workplace mechanisms.

References

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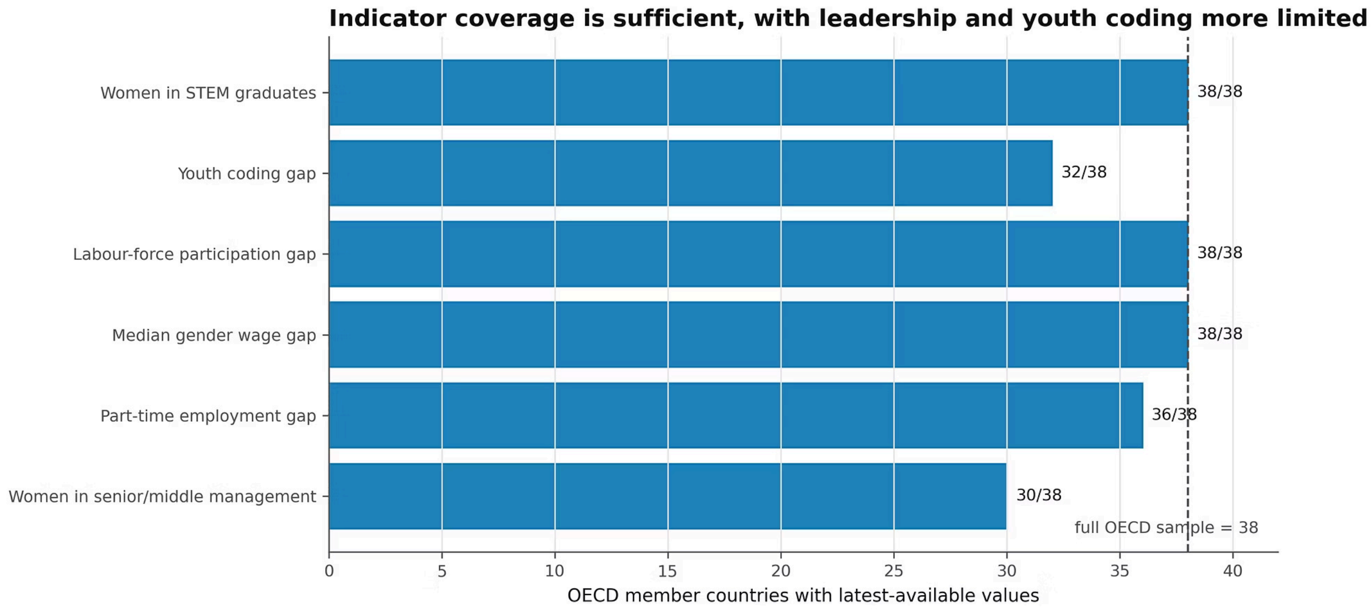
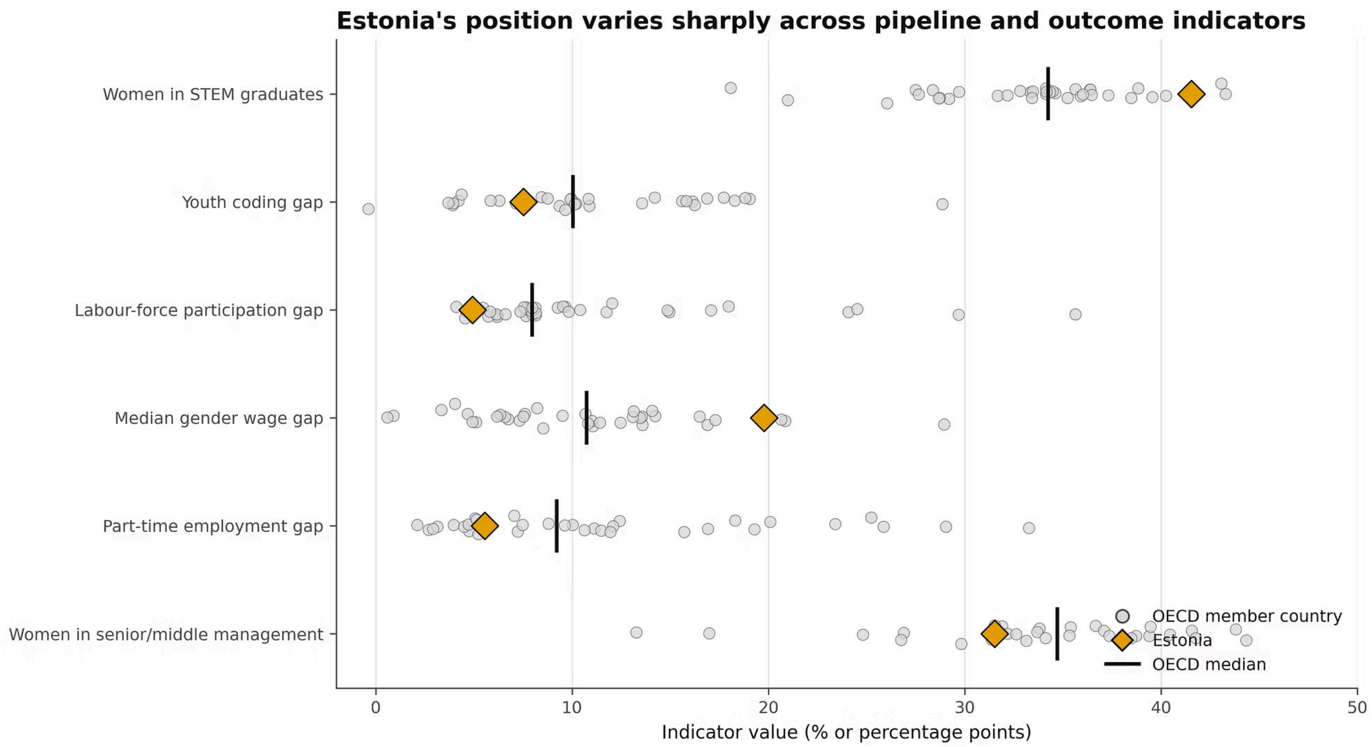
The analysis notebooks, processed reference tables, and report figures are available in the project repository: **https://github.com/kelukes/gender-equality-statistical-report**

Appendix A — Supplementary Data Coverage and Benchmark Figures

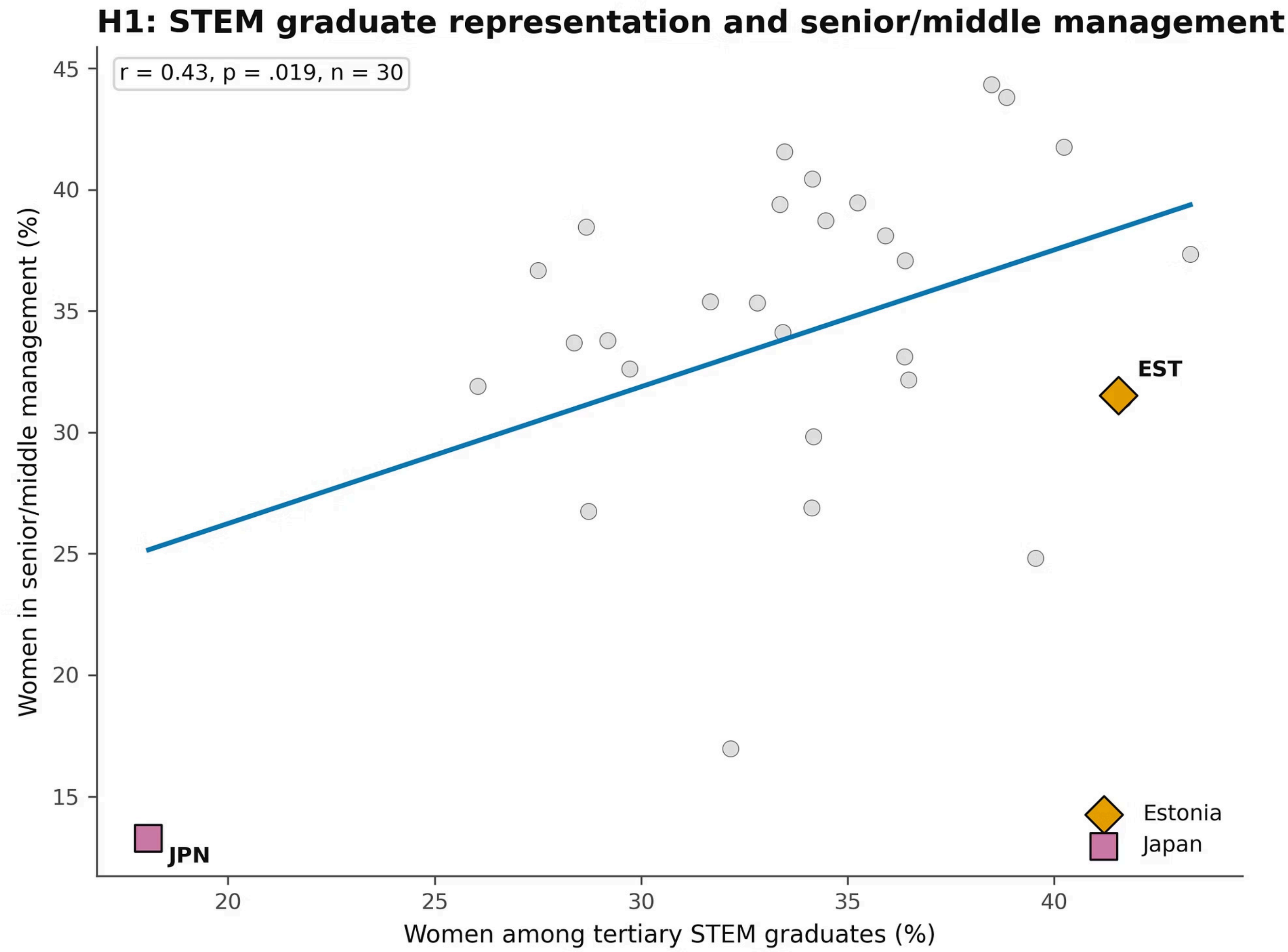
These figures support the data quality and descriptive benchmarking sections.

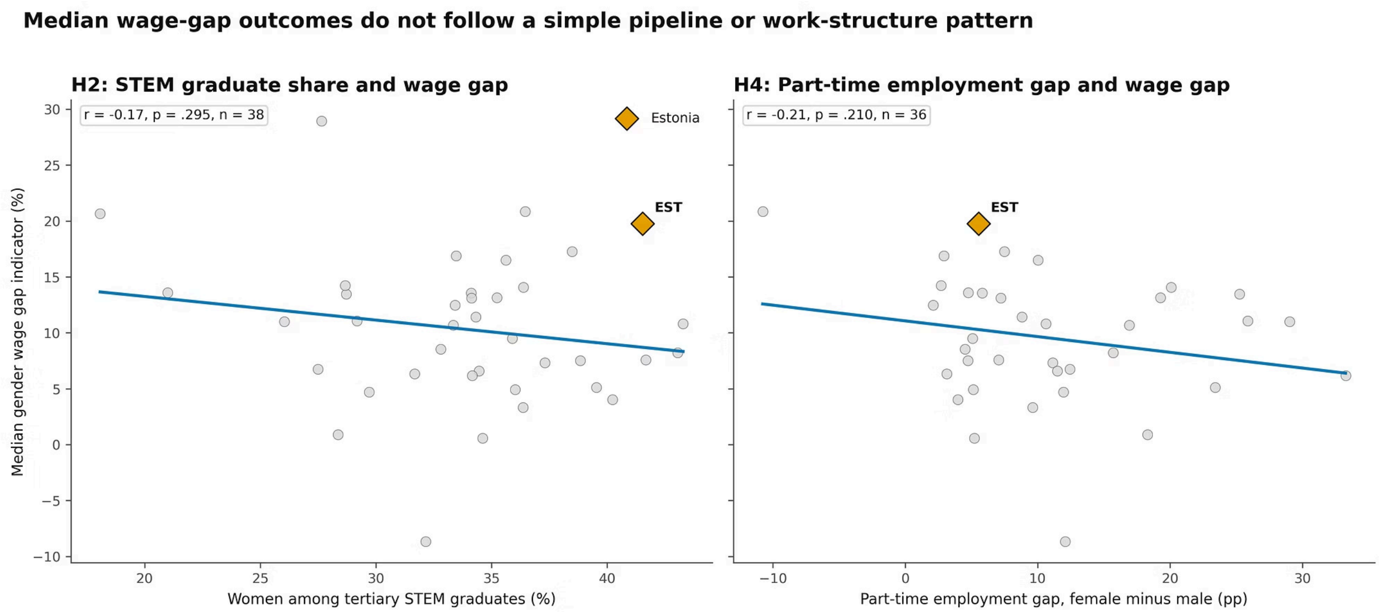




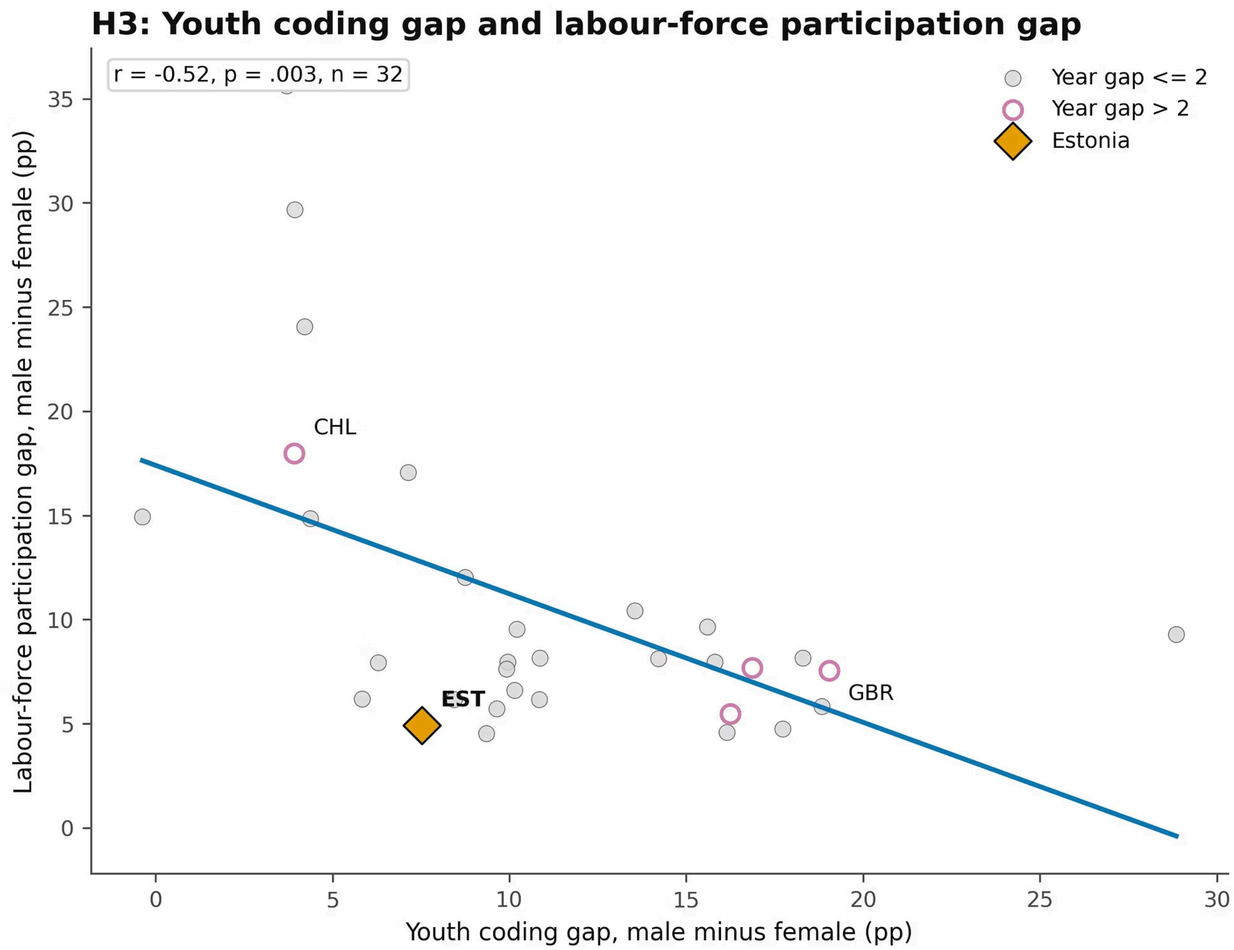
Appendix B — Supplementary Hypothesis Diagnostics

These plots show the pairwise relationships behind the hypothesis tests and support the interpretation of partial, unsupported, and exploratory findings.





Appendix B continued — H3 exploratory diagnostic



Exploratory only: youth coding covers ages 16-24, while labour-force participation covers ages 15-74.